

$$\text{If } |z| = 1$$

$$|z^2 + z + 1| < |z|^2 + |z| + 1 = 3$$

$$|z^3 - 2| > ||z|^3 - |2|| = |1 - 2| = 1$$

$$\text{If } |z| = 1/2$$

$$\left| \frac{1}{z^2 - 5z + 6} \right| \leq \frac{4}{15}$$

$$\left| \frac{1}{z^2 - 5z + 6} \right| = \left| \frac{1}{(z-3)(z-2)} \right| = \frac{1}{|z-3||z-2|}$$

$$||z| - |w|| < |z - w|$$

$$|z-3| >, ||z| - |3|| = |1/2 - 3| = 5/2$$

$$|z-2| >, ||z| - |2|| = |1/2 - 2| = 3/2$$

$$\frac{1}{|z-3|} < \frac{2}{5} \quad \frac{1}{|z-2|} < \frac{2}{3}$$

$$\frac{1}{|z-3||z-2|} < \frac{2}{5} \cdot \frac{2}{3} = \frac{4}{15}$$

if $z = \frac{i}{3+i}$, $\bar{z} = ?$ $|z| = ?$

$$z = \frac{i}{3+i} = \frac{i(3-i)}{(3+i)(3-i)} = \frac{3i+1}{10} = \frac{1}{10} + i\frac{3}{10}$$

$$z = \frac{1}{10} + \frac{3i}{10} \Rightarrow \bar{z} = \frac{1}{10} - \frac{3i}{10}$$

$$|z| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(-\frac{3}{10}\right)^2} = \sqrt{\frac{10}{100}} = \frac{\sqrt{10}}{10} = \frac{1}{\sqrt{10}}$$

Show that the following equations are true.

a) $(1-i)^4 = -4$

b) $\frac{5}{(1-i)(2-i)(3-i)} = \frac{i}{2}$

c) $\overline{(\bar{z} + 3i)} = z - 3$

d) $\operatorname{Re}(iz) = -\operatorname{Im} z$

if $|z|=2$ $\left| \frac{-1}{z^4 + 3z^2 + 2} \right|$

Find an upper bound

$$\frac{|-1|}{|z^4 + 3z^2 + 2|} = \frac{1}{|z^4 + 3z^2 + 2|} \leq M \quad M = ?$$

$$|z^4 + 3z^2 + 2| = |z^2 + 1| |z^2 + 2| \geq ||z^2| - 1| \cdot |z^2 + 2|$$

$$||z_1| - |z_2|| \leq |z_1 + z_2| \quad |z|=2$$

$$|z^2 + 3z^2 + 2| \geq 3 \cdot 2 = 6$$

$$\frac{1}{|z^2 + 3z^2 + 2|} \leq \frac{1}{6}$$

Ex: Let $\arg w$ is defined $-\pi \leq w \leq \pi$ and z is on $|z|=1$.

show that $\arg \frac{z-1}{z+1} = \begin{cases} \pi/2 & \text{if } \operatorname{Im} z > 0 \\ -\pi/2 & \text{if } \operatorname{Im} z < 0 \end{cases}$

Let $z = x + iy$; $x, y \in \mathbb{R}$; on unit circle.

$$\begin{aligned} \frac{z-1}{z+1} &= \frac{x+iy-1}{x+iy+1} = \frac{(x-1)+iy}{(x+1)+iy} = \frac{[(x-1)+iy][(x+1)-iy]}{[(x+1)+iy][(x+1)-iy]} \\ &= \frac{(x-1)(x+1) - (x-1) \cdot y i + (x+1) \cdot iy + y^2}{(x+1)^2 + y^2} \\ &= \frac{\cancel{x^2} - 1 + ((x+1)y - (x-1)y)i + y^2}{(x+1)^2 + y^2} \\ &\quad \downarrow \\ &\quad x^2 + y^2 = 1 \end{aligned}$$

$$x^2 + y^2 = 1$$

$$\frac{z-1}{z+1} = \frac{2yi}{(x+1)^2 + y^2}$$

$$\Rightarrow \arg \left(\frac{z-1}{z+1} \right) = \begin{cases} \frac{\pi}{2}; & y > 0 \\ -\frac{\pi}{2}; & y < 0 \end{cases}$$

LET US REMEMBER

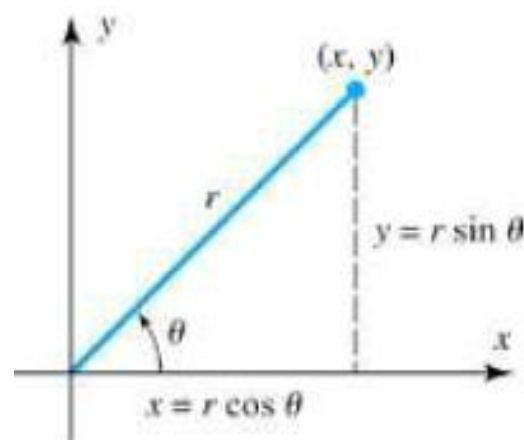
Polar Form

If $z = x + iy$ is a nonzero complex number, $r = |z|$, and θ measures the angle from the positive real axis to the vector z , then, as suggested by Figure

$$x = r \cos \theta, \quad y = r \sin \theta$$

so that $z = x + iy$ can be written as $z = r \cos \theta + ir \sin \theta$ or

$$z = r(\cos \theta + i \sin \theta)$$



This is called a *polar form of z*.

Argument of a Complex Number

The angle θ is called an *argument of z* and is denoted by

$$\theta = \arg z$$

The argument of z is not uniquely determined because we can add or subtract any multiple of 2π from θ to produce another value of the argument. However, there is only one value of the argument in radians that satisfies

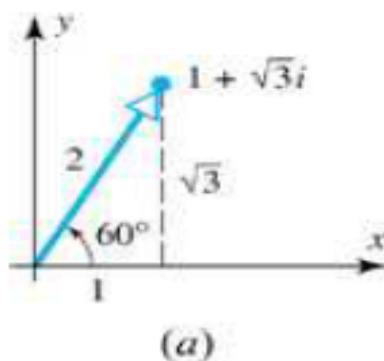
$$-\pi < \theta \leq \pi$$

This is called the *principal argument of z* and is denoted by

$$\theta = \text{Arg } z$$

Express the following complex numbers in polar form using their principal arguments:

(a) $z = 1 + \sqrt{3}i$



The value of r is

$$r = |z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

and since $x = 1$ and $y = \sqrt{3}$, it follows from 1 that

$$1 = 2 \cos \theta \quad \text{and} \quad \sqrt{3} = 2 \sin \theta$$

so $\cos \theta = 1/2$ and $\sin \theta = \sqrt{3}/2$. The only value of θ that satisfies these relations and meets the requirement $-\pi < \theta \leq \pi$ is $\theta = \pi/3 (= 60^\circ)$. Thus a polar form of z is

$$z = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

(b) $z = -1 - i$

The value of r is

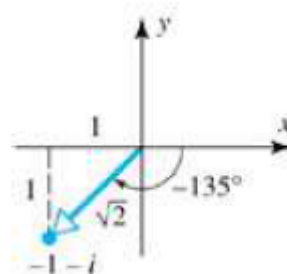
$$r = |z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

and since $x = -1$, $y = -1$, it follows from 1 that

$$-1 = \sqrt{2} \cos \theta \quad \text{and} \quad -1 = \sqrt{2} \sin \theta$$

so $\cos \theta = -1 / \sqrt{2}$ and $\sin \theta = -1 / \sqrt{2}$. The only value of θ that satisfies these relations and meets the requirement $-\pi < \theta \leq \pi$ is $\theta = -3\pi / 4 (= -135^\circ)$. Thus, a polar form of z is

$$z = \sqrt{2} \left(\cos \frac{-3\pi}{4} + i \sin \frac{-3\pi}{4} \right)$$



(b)

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1) \quad \text{and} \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$$

$$z_1 z_2 = r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)]$$

$$\cos(\theta_1 + \theta_2) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2$$

$$\sin(\theta_1 + \theta_2) = \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2$$

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

if $z_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

Let

$$z_1 = 1 + \sqrt{3}i \quad \text{and} \quad z_2 = \sqrt{3} + i$$

Polar forms of these complex numbers are

$$z_1 = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \quad \text{and} \quad z_2 = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$\begin{aligned} z_1 z_2 &= 4 \left[\cos \left(\frac{\pi}{3} + \frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{3} + \frac{\pi}{6} \right) \right] \\ &= 4 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right] = 4[0 + i] = 4i \end{aligned}$$

$$\begin{aligned} \frac{z_1}{z_2} &= 1 \cdot \left[\cos \left(\frac{\pi}{3} - \frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \right] \\ &= \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i \end{aligned}$$

As a check, we calculate $z_1 z_2$ and z_1 / z_2 directly without using polar forms for z_1 and z_2 :

$$\begin{aligned} z_1 z_2 &= (1 + \sqrt{3}i)(\sqrt{3} + i) = (\sqrt{3} - \sqrt{3}) + (3 + 1)i = 4i \\ \frac{z_1}{z_2} &= \frac{1 + \sqrt{3}i}{\sqrt{3} + i} \cdot \frac{\sqrt{3} - i}{\sqrt{3} - i} = \frac{(\sqrt{3} + \sqrt{3}) + (-i + 3i)}{4} = \frac{\sqrt{3}}{2} + \frac{1}{2}i \end{aligned}$$

which agrees with our previous results.

DeMoivre's Formula

If n is a positive integer and $z = r(\cos \theta + i \sin \theta)$, then

$$z^n = \underbrace{z \cdot z \cdot z \cdots z}_{n\text{-factors}} = r^n \left[\underbrace{\cos(\theta + \theta + \cdots + \theta)}_{n\text{-terms}} + i \underbrace{\sin(\theta + \theta + \cdots + \theta)}_{n\text{-terms}} \right]$$

or

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$

In the special case where $r = 1$, we have $z = \cos \theta + i \sin \theta$,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

which is called *DeMoivre's formula*.

Finding n th Roots

We now show how DeMoivre's formula can be used to obtain roots of complex numbers. If n is a positive integer and z is any complex number, then we define an *n th root of z* to be any complex number w that satisfies the equation

$$w^n = z \quad *$$

We denote an n th root of z by $z^{1/n}$. If $z \neq 0$, then we can derive formulas for the n th roots of z as follows. Let

$$w = \rho(\cos \alpha + i \sin \alpha) \quad \text{and} \quad z = r(\cos \theta + i \sin \theta)$$

$$\rho^n(\cos n\alpha + i \sin n\alpha) = r(\cos \theta + i \sin \theta)$$

Comparing the moduli of the two sides, we see that $\rho^n = r$ or

$$\rho = \sqrt[n]{r}$$

where $\sqrt[n]{r}$ denotes the real positive n th root of r . Moreover, in order to have the equalities $\cos n\alpha = \cos \theta$ and $\sin n\alpha = \sin \theta$ the angles $n\alpha$ and θ must either be equal or differ by a multiple of 2π . That is,

$$n\alpha = \theta + 2k\pi \quad \text{or} \quad \alpha = \frac{\theta}{n} + \frac{2k\pi}{n}, \quad k = 0, \pm 1, \pm 2, \dots$$

Thus the values of $w = \rho(\cos \alpha + i \sin \alpha)$ that satisfy $z = w^n$ are given by

$$w = \sqrt[n]{r} \left[\cos \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) \right], \quad k = 0, \pm 1, \pm 2, \dots$$

Although there are infinitely many values of k , it can be shown (see Exercise 10) that $k = 0, 1, 2, \dots, n-1$ produce distinct values of w satisfying $z = w^n$ but all other choices of k yield duplicates of these. Therefore, there are exactly n different n th roots of $z = r(\cos \theta + i \sin \theta)$, and these are given by

$$z^{1/n} = \sqrt[n]{r} \left[\cos \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) \right], \quad k = 0, 1, 2, \dots, n-1$$

Find the three cube roots of $z = i$.

Solution Keep in mind that we are basically solving the equation $w^3 = i$. Now with $r = 1$, $\theta = \arg(i) = \pi/2$, a polar form of the given number is given by $z = \cos(\pi/2) + i \sin(\pi/2)$.] with $n = 3$, we then obtain

$$w_k = \sqrt[3]{1} \left[\cos \left(\frac{\pi/2 + 2k\pi}{3} \right) + i \sin \left(\frac{\pi/2 + 2k\pi}{3} \right) \right], \quad k = 0, 1, 2.$$

Hence the three roots are,

$$k = 0, \quad w_0 = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 1, \quad w_1 = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 2, \quad w_2 = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} = -i.$$

Find all cube roots of -8 .

Solution

Since -8 lies on the negative real axis, we can use $\theta = \pi$ as an argument. Moreover, $r = |z| = |-8| = 8$, so a polar form of -8 is

$$-8 = 8(\cos \pi + i \sin \pi)$$

$n = 3$, it follows that

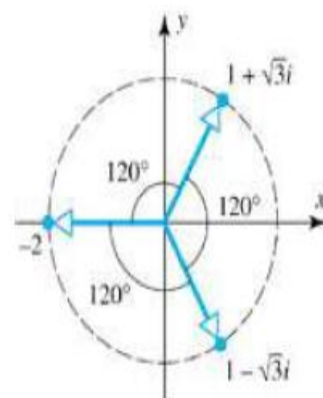
$$(-8)^{1/3} = \sqrt[3]{8} \left[\cos \left(\frac{\pi}{3} + \frac{2k\pi}{3} \right) + i \sin \left(\frac{\pi}{3} + \frac{2k\pi}{3} \right) \right], \quad k = 0, 1, 2$$

Thus the cube roots of -8 are

$$2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) = 1 + \sqrt{3}i$$

$$2(\cos \pi + i \sin \pi) = 2(-1) = -2$$

$$2 \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right) = 2 \left(\frac{1}{2} - \frac{\sqrt{3}}{2} i \right) = 1 - \sqrt{3}i$$



The cube roots of -8 .

Fourth Roots of a Complex Number

Find the four fourth roots of $z = 1 + i$.

Solution In this case, $r = \sqrt{2}$ and $\theta = \arg(z) = \pi/4$.
we obtain

$$n = 4,$$

$$w_k = (\sqrt{2})^{1/4} \left[\cos \left(\frac{\pi/4 + 2k\pi}{4} \right) + i \sin \left(\frac{\pi/4 + 2k\pi}{4} \right) \right], \quad k = 0, 1, 2, 3.$$

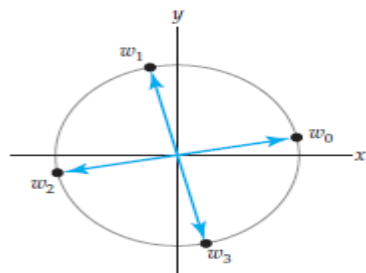
With the aid of a calculator we find

$$k = 0, \quad w_0 = (\sqrt{2})^{1/4} \left[\cos \frac{\pi}{16} + i \sin \frac{\pi}{16} \right] \approx 1.1664 + 0.2320i$$

$$k = 1, \quad w_1 = (\sqrt{2})^{1/4} \left[\cos \frac{9\pi}{16} + i \sin \frac{9\pi}{16} \right] \approx -0.2320 + 1.1664i$$

$$k = 2, \quad w_2 = (\sqrt{2})^{1/4} \left[\cos \frac{17\pi}{16} + i \sin \frac{17\pi}{16} \right] \approx -1.1664 - 0.2320i$$

$$k = 3, \quad w_3 = (\sqrt{2})^{1/4} \left[\cos \frac{25\pi}{16} + i \sin \frac{25\pi}{16} \right] \approx 0.2320 - 1.1664i.$$



3 Four fourth roots of $1 + i$

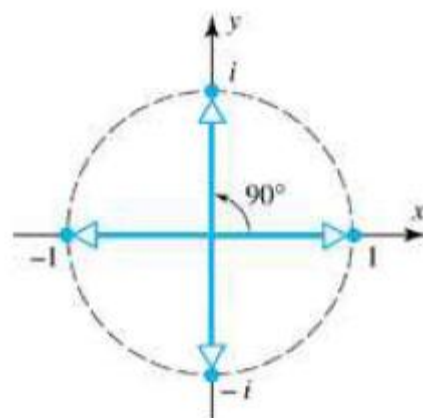
Geometrically, the n n th roots of a complex number z can also be interpreted as the vertices of a regular polygon with n sides that is inscribed within a circle of radius $\sqrt[n]{r}$ centered at the origin.

Find all fourth roots of 1.

Solution

$w = 1$ is one fourth root of 1, so the remaining three roots can be generated by rotating this root through increments of $2\pi / 4 = \pi / 2$ radians ($= 90^\circ$).
roots of 1 are

$$1, \quad i, \quad -1, \quad -i$$



$$z_1 = r_1 e^{i\theta_1} \quad \text{and} \quad z_2 = r_2 e^{i\theta_2}$$

are nonzero complex numbers, then

$$z_1 z_2 = r_1 r_2 e^{i\theta_1 + i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i\theta_1 - i\theta_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

We conclude this section with a useful formula for $\bar{z}$ in polar notation. If

$$z = r e^{i\theta} = r(\cos \theta + i \sin \theta)$$

then

$$\bar{z} = r(\cos \theta - i \sin \theta)$$

Recalling the trigonometric identities

$$\sin(-\theta) = -\sin \theta \quad \text{and} \quad \cos(-\theta) = \cos \theta$$

$$\bar{z} = r[\cos(-\theta) + i \sin(-\theta)] = r e^{i(-\theta)}$$

or, equivalently,

$$\bar{z} = r e^{-i\theta}$$

In the special case where $r = 1$, the polar form of z is $z = e^{i\theta}$,

$$\overline{e^{i\theta}} = e^{-i\theta}$$

$$az^2 + bz + c = 0 \quad a, b, c \in \mathbb{R}$$

SOME EX.

$$a\left(z^2 + \frac{b}{a}z\right) + c = 0$$

$$a\left(\left(z + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right) + c = 0 \Rightarrow a\left(z + \frac{b}{2a}\right)^2 = \frac{b^2}{4a} - c$$

$$a\left(z + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a} \quad \checkmark$$

$$z + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$z = -\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$b^2 - 4ac < 0 \Rightarrow z = \frac{-b \pm \sqrt{\Delta} \sqrt{-1}}{2a} = \frac{-b \pm \sqrt{\Delta} i}{2a}$$

$$(2) \quad \overset{a}{i} z^2 + \overset{b}{(1+i)} z - \overset{c}{2} i = 0$$

$$z^2 + \frac{(1+i)}{i} z - 2 = 0 \Rightarrow \overset{a}{1} z^2 + \overset{b}{(1-i)} z - \overset{c}{2} = 0$$

$$\left(z + \frac{(1-i)}{2} \right)^2 = 2 - \frac{i}{2}$$

⇓

$$\left| 2 - \frac{i}{2} \right| = \sqrt{4 + \frac{1}{4}} = \frac{\sqrt{17}}{2}$$

$$\text{Arg}\left(2 - \frac{i}{2}\right) = \theta = \arctan\left(\frac{-1/2}{2}\right)$$

$$= \arctan(-1/4) = -14.036^\circ$$

$$z_k = -\frac{1+i}{2} + \left(\frac{\sqrt{17}}{2}\right)^{1/2} \left(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2} \right)$$

$$k = 0, 1$$

$$(3) \quad z = (1+i)^n + (1-i)^n$$

$$|1+i| = \sqrt{2} \quad \text{Arg}(1+i) = \frac{\pi}{4}$$

$$(1+i) = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$(1+i)^n = (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)$$

$$(1-i)^n = (\sqrt{2})^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right)$$

$$z = 2(\sqrt{2})^n \left(\cos \frac{n\pi}{4} \right) = \bar{z}$$

$$|z| = 2(\sqrt{2})^n \left| \cos \frac{n\pi}{4} \right|$$

show that

(4) $z, w \in \mathbb{C} \Rightarrow |z+w|^2 + |z-w|^2 = 2(|z|^2 + |w|^2)$

$$|z+w|^2 = (z+w)(\overline{z+w}) = z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w}$$

$$|z-w|^2 = (z-w)(\overline{z-w}) = z\bar{z} - z\bar{w} - w\bar{z} + w\bar{w}$$

$$= 2(|z|^2 + |w|^2)$$

(5) for $a \in \mathbb{C}$; $\downarrow$ constant

if $1 - \bar{a}z \neq 0$ and $|z|=1$; $\left| \frac{z-a}{1-\bar{a}z} \right| = 1$

$$\frac{|z-a|}{|1-\bar{a}z|} = \frac{|z-a|}{\underbrace{|\bar{z}|}_{|z|=1} |1-\bar{a}z|} = \frac{|z-a|}{|\bar{z}(1-\bar{a}z)|} = \frac{|z-a|}{(\bar{z}-\bar{a}|z|^2)} = \frac{|z-a|}{|\bar{z}-\bar{a}|}$$

$$= \frac{|z-a|}{|z-a|} = 1$$

$$z^{2/3} = (1 - \sqrt{3}i)$$

$$z^2 = (1 - \sqrt{3}i)^3$$

$$2\pi - \frac{\pi}{3}$$

$$1 - \sqrt{3}i = 2 \cdot \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right)$$

$$r = \sqrt{1^2 + (\sqrt{3})^2}$$

$$r = 2$$

$$\text{Arg}(1 - \sqrt{3}i) = \theta = -\frac{\pi}{3}$$



$$2\pi + \frac{\pi}{2} = \frac{5\pi}{2}$$

$$3\pi + \frac{\pi}{2} = \frac{7\pi}{2}$$

$$\Rightarrow z^2 = (1 - \sqrt{3}i)^3 = (2)^3 \left(\cos 3 \cdot \frac{5\pi}{3} + i \sin 3 \cdot \frac{5\pi}{3} \right)$$

$$z^2 = 8 \cdot (\cos 5\pi + i \sin 5\pi)$$

$$z_k = 8^{1/2} \left(\cos \frac{5\pi + 2k\pi}{2} + i \sin \frac{5\pi + 2k\pi}{2} \right)$$

$$z_0 = 2\sqrt{2}i$$

$$z_1 = -2\sqrt{2}i$$

$$\underline{\underline{k=0, 1}}$$

$$|\alpha| < 1 \Rightarrow |z| \leq 1 \Leftrightarrow$$

$$\frac{|z - \alpha|}{|1 - \bar{\alpha}z|} \leq 1$$

$$\left| \frac{z - \alpha}{1 - \bar{\alpha}z} \right| \leq 1 \Leftrightarrow |z - \alpha| \leq |1 - \bar{\alpha}z|$$

$$|z - \alpha|^2 \leq |1 - \bar{\alpha}z|^2$$

$$(z - \alpha)(\bar{z} - \bar{\alpha}) \leq (1 - \bar{\alpha}z)(1 - \alpha\bar{z})$$

$$(z - \alpha)(\bar{z} - \bar{\alpha}) \leq (1 - \bar{\alpha}z)(1 - \alpha\bar{z})$$

$$z\bar{z} - z\bar{\alpha} - \alpha\bar{z} + \alpha\bar{\alpha} \leq 1 - \alpha\bar{z} - \bar{\alpha}z + \bar{\alpha}\alpha\bar{z}z$$

$$|z|^2 + |\alpha|^2 \leq 1 + |\alpha|^2|z|^2$$

$$|z|^2 - |\alpha|^2|z|^2 + |\alpha|^2 - 1 \leq 0$$

$$|z|^2 [1 - |\alpha|^2] + [|\alpha|^2 - 1] \leq 0$$

$$(|\alpha|^2 - 1)(1 - |z|^2) \leq 0 \quad \text{because } |\alpha| < 1$$

$$|\alpha|^2 - 1 < 0$$

$$\Downarrow$$

$$\leq 1$$

Circles Suppose $z_0 = x_0 + iy_0$. Since $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$ is the distance between the points $z = x + iy$ and $z_0 = x_0 + iy_0$, the points $z = x + iy$ that satisfy the equation

$$|z - z_0| = \rho, \quad \rho > 0, \tag{1}$$

lie on a **circle** of radius ρ centered at the point z_0 .

- (a) $|z| = 1$ is an equation of a unit circle centered at the origin.
- (b) By rewriting $|z - 1 + 3i| = 5$ as $|z - (1 - 3i)| = 5$, we see from (1) that the equation describes a circle of radius 5 centered at the point $z_0 = 1 - 3i$.

$$|z - 4 + 3i| = 5$$

$$z = a + ib ; 0, b \in \mathbb{R}$$

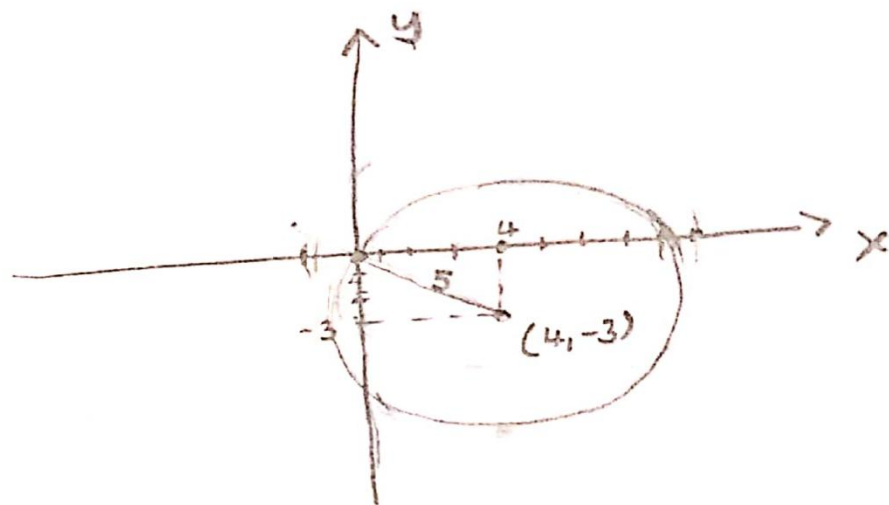
$$|a + ib - 4 + 3i| = 5$$

$$|a - 4 + i(b + 3)| = 5$$

$$\sqrt{(a - 4)^2 + (b + 3)^2} = 5$$

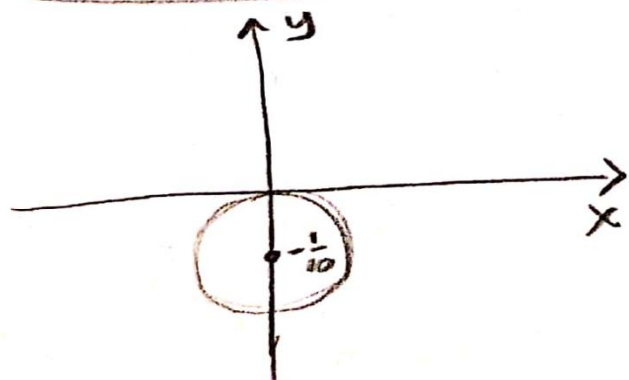
$$(a - 4)^2 + (b + 3)^2 = 25$$

$$\text{center: } (4, -3) \quad r = 5$$



$$\operatorname{Im} \left(\frac{1}{z} \right) = 5 \Rightarrow \operatorname{Im} \left(\frac{1}{x + iy} \right) = \operatorname{Im} \left(\frac{x - iy}{(x + iy)(x - iy)} \right) = \frac{-y}{x^2 + y^2} = 5$$

$$z = x + iy, \quad x, y \in \mathbb{R}$$



$$-y = 5x^2 + 5y^2$$

$$5x^2 + 5y^2 + y = 0$$

$$5 \left(x^2 + y^2 + \frac{y}{5} \right) = 0$$

$$x^2 + \left(y + \frac{1}{10} \right)^2 - \frac{1}{100} = 0$$

$$x^2 + \left(y + \frac{1}{10} \right)^2 = \frac{1}{100}$$

$$\text{center: } (0, -\frac{1}{10}) \quad r = \frac{1}{10}$$

if $\left| \frac{z-4}{z+4} \right| \leq 3$; $z = a+ib$; $a, b \in \mathbb{R}$

$$\left| \frac{a+ib-4}{a+ib+4} \right| \leq 3 \Rightarrow |(a-4)+ib| \leq 3|(a+4)+ib|$$

$$\sqrt{(a-4)^2+b^2} \leq 3\sqrt{(a+4)^2+b^2}$$

$$(a-4)^2+b^2 \leq 9[(a+4)^2+b^2]$$

$$a^2-8a+16+b^2 \leq 9a^2+72a+9b^2+9b^2$$

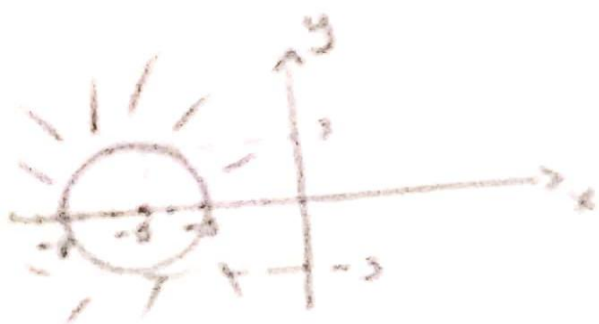
$$0 \leq 8a^2+80a+8b^2+8b^2$$

$$0 \leq a^2+10a+1b+b^2$$

$$0 \leq (a+5)^2+b^2-9$$

$$9 \leq (a+5)^2+b^2$$

center : $(-5, 0)$
 $r = 3$



if $|z-i| > 1$

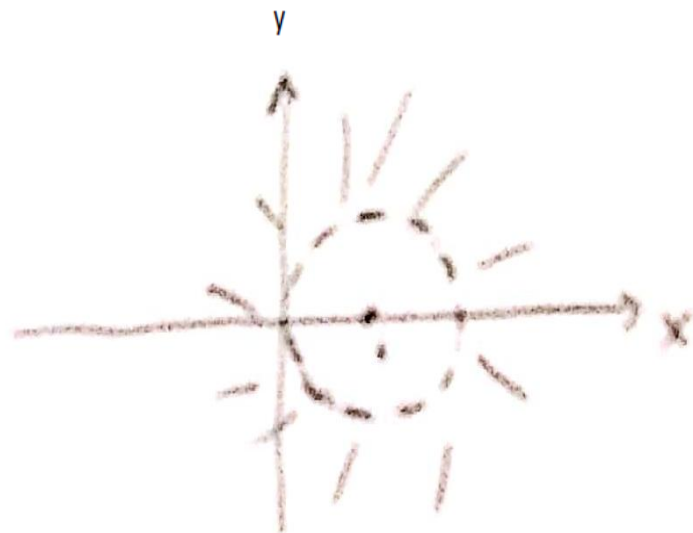
$$|a+ib-i| > 1$$

$$|a+i(b-1)| > 1$$

$$\sqrt{a^2 + (b-1)^2} > 1$$

$$a^2 + (b-1)^2 > 1$$

Center: $(0,1)$ $r=1$



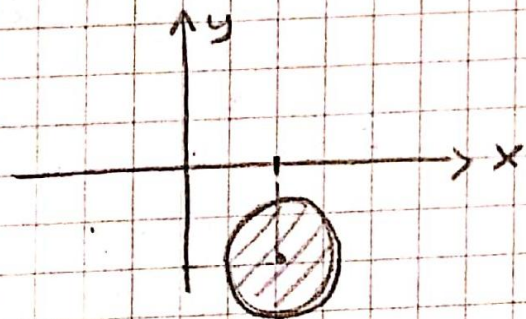
$$|z - 1 + i| \leq 1/2$$

$$|z - (1 - i)| \leq 1/2$$

$$|(x-1) - i(y+1)| \leq 1/2$$

$$(x-1)^2 + (y+1)^2 \leq 1/4$$

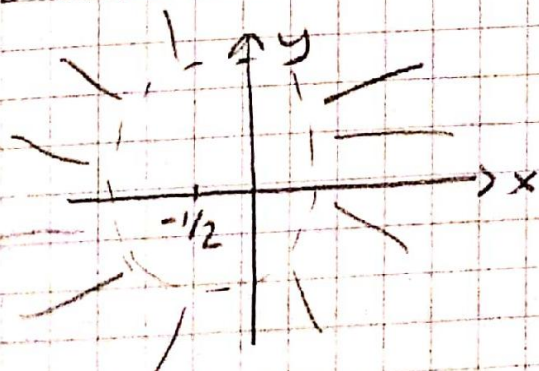
$(1, -1) \rightarrow$ center, $r = 1/2 \rightarrow$ radius



$$|2z + 1| > 2$$

$$|2z + 1| = 2 \Rightarrow |z + 1/2| > 1 \quad (-1/2, 0)$$

$$r = 1$$

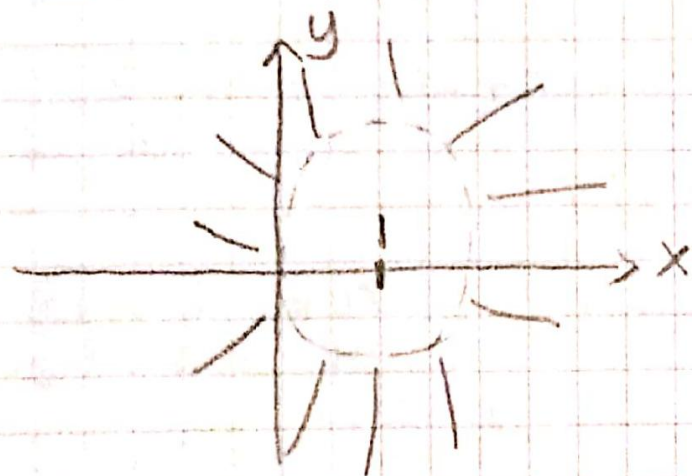


$$2 \operatorname{Re} z < |z|^2$$

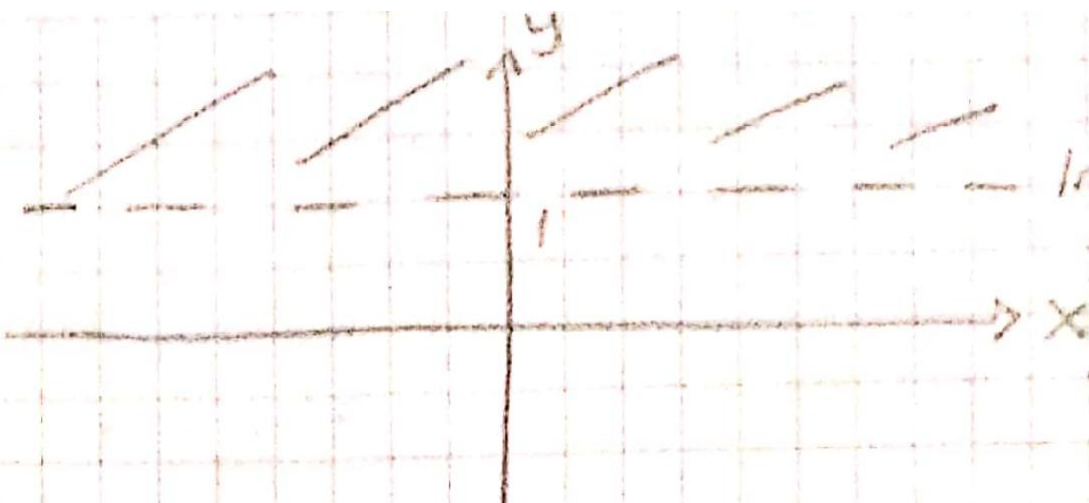
$$2x < x^2 + y^2 \Rightarrow 0 < x^2 - 2x + y^2 + 1 - 1$$

$$1 < (x-1)^2 + y^2$$

$(1, 0) \rightarrow$ center
 $r = 1 \rightarrow$ radius



$$|\operatorname{Im} z| > 1$$



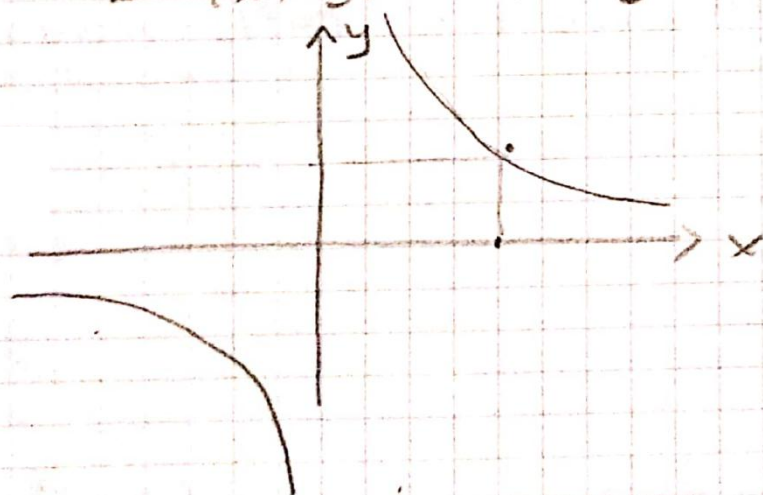
$$|\operatorname{Im} z| = 1$$

$$\operatorname{Im}(z^2) = 4$$

$$z^2 = (x+iy)^2$$

$$x^2 - y^2 + 2ixy \Rightarrow \operatorname{Im} z^2 = 2xy = 4$$

$$xy = 2$$



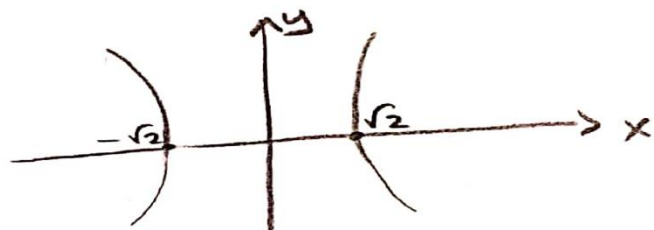
$$\operatorname{Re}(z^2) = |\sqrt{3} - i|$$

$$z = a + ib, \quad a, b \in \mathbb{R}$$

$$\operatorname{Re}((a+ib)^2) = \operatorname{Re}(a^2 + 2iab - b^2) = |\sqrt{3} - i|$$

$$a^2 - b^2 = 2 \Rightarrow \frac{a^2}{(\sqrt{2})^2} - \frac{b^2}{(\sqrt{2})^2} = 1$$

hyperbola



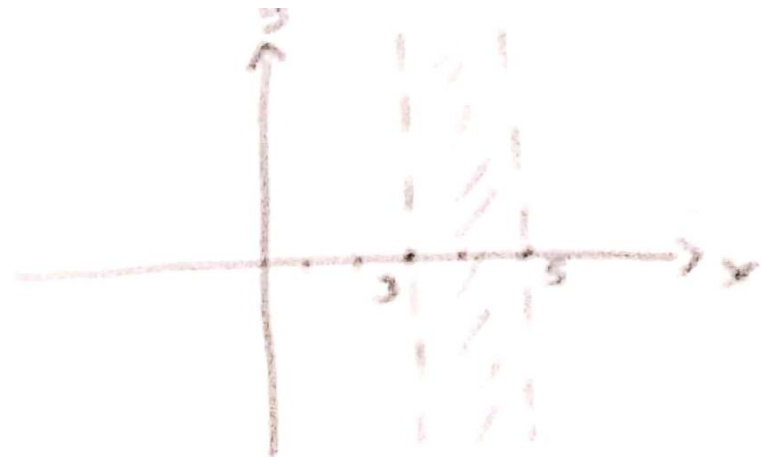
if

$$2 < \operatorname{Re}(z-1) < 4$$

$$2 < \operatorname{Re}(x+iy-1) < 4$$

$$2 < x-1 < 4$$

$$3 < x < 5$$



if $|z-i| > 1$

$$|a+ib-i| > 1$$

$$|a+i(b-1)| > 1$$

$$\sqrt{a^2 + (b-1)^2} > 1$$

$$a^2 + (b-1)^2 > 1$$

Center: (0,1) $r=1$



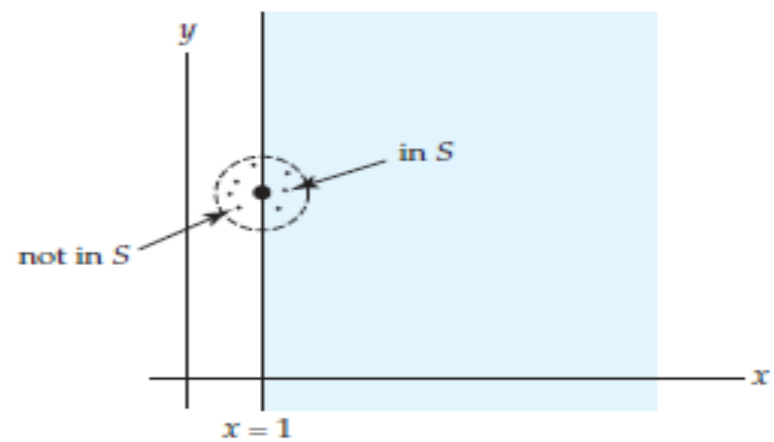
Disks and Neighborhoods

The points z that satisfy the inequality $|z - z_0| \leq \rho$ can be either *on* the circle $|z - z_0| = \rho$ or *within* the circle. We say that the set of points defined by $|z - z_0| \leq \rho$ is a **disk** of radius ρ centered at z_0 . But the points z that satisfy the strict inequality $|z - z_0| < \rho$ lie within, and not on, a circle of radius ρ centered at the point z_0 . This set is called a **neighborhood** of z_0 . Occasionally, we will need to use a neighborhood of z_0 that also *excludes* z_0 . Such a neighborhood is defined by the simultaneous inequality $0 < |z - z_0| < \rho$ and is called a **deleted neighborhood** of z_0 . For example, $|z| < 1$ defines a neighborhood of the origin, whereas $0 < |z| < 1$ defines a deleted neighborhood of the origin; $|z - 3 + 4i| < 0.01$ defines a neighborhood of $3 - 4i$, whereas the inequality $0 < |z - 3 + 4i| < 0.01$ defines a deleted neighborhood of $3 - 4i$.

Open Sets

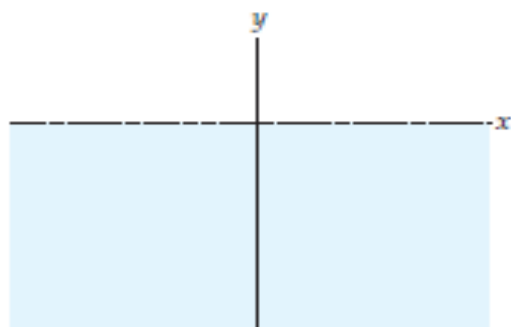
A point z_0 is said to be an **interior point** of a set S of the complex plane if there exists some neighborhood of z_0 that lies entirely within S . If every point z of a set S is an interior point, then S is said to be an **open set**.

For example, the inequality $\operatorname{Re}(z) > 1$ defines a *right half-plane*, which is an open set.

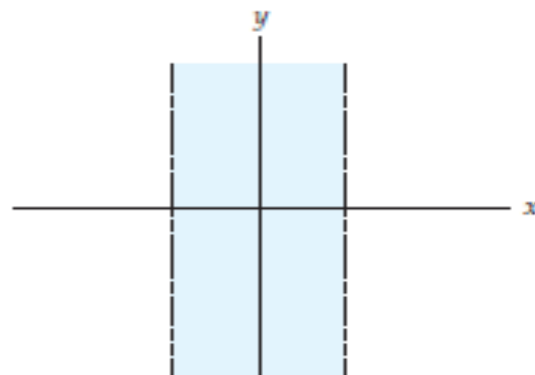


| Set S not open

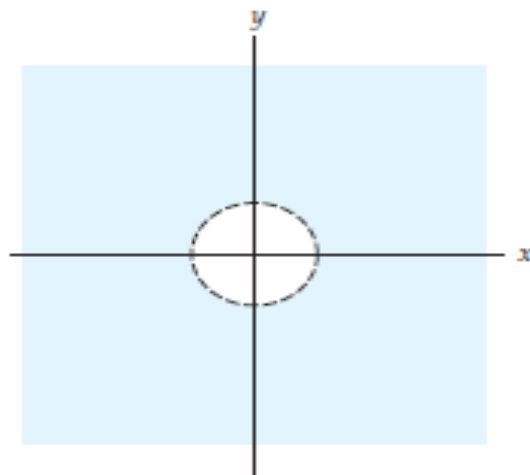
Some Open Sets



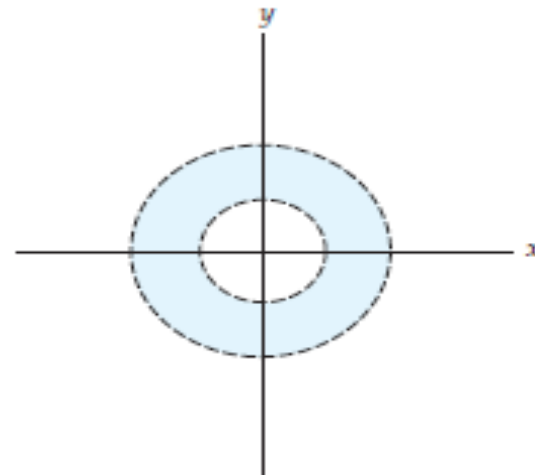
(a) $\text{Im}(z) < 0$; lower half-plane



(b) $-1 < \text{Re}(z) < 1$; infinite vertical strip

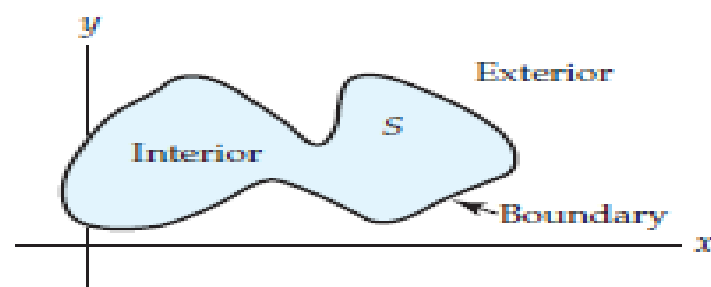


(c) $|z| > 1$; exterior of unit circle



(d) $1 < |z| < 2$; interior of circular ring

If *every* neighborhood of a point z_0 of a set S contains at least one point of S and at least one point not in S , then z_0 is said to be a **boundary point** of S . For the set of points defined by $\operatorname{Re}(z) \geq 1$, the points on the vertical line $x = 1$ are boundary points. The points that lie on the circle $|z - i| = 2$ are boundary points for the disk $|z - i| \leq 2$ as well as for the neighborhood $|z - i| < 2$ of $z = i$. The collection of boundary points of a set S is called the **boundary** of S . The circle $|z - i| = 2$ is the boundary for both the disk $|z - i| \leq 2$ and the neighborhood $|z - i| < 2$ of $z = i$. A point z that is neither an interior point nor a boundary point of a set S is said to be an **exterior point** of S ; in other words, z_0 is an exterior point of a set S if there exists *some* neighborhood of z_0 that contains no points of S .

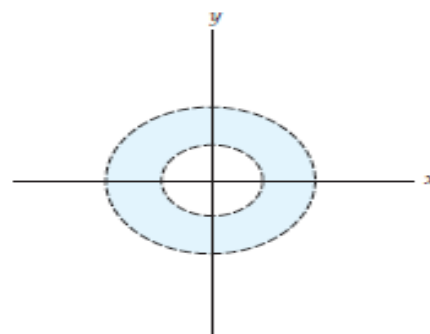


Annulus

The set S_1 of points satisfying the inequality $\rho_1 < |z - z_0|$ lie exterior to the circle of radius ρ_1 centered at z_0 , whereas the set S_2 of points satisfying $|z - z_0| < \rho_2$ lie interior to the circle of radius ρ_2 centered at z_0 . Thus, if $0 < \rho_1 < \rho_2$, the set of points satisfying the simultaneous inequality

$$\rho_1 < |z - z_0| < \rho_2, \quad (2)$$

is the intersection of the sets S_1 and S_2 . This intersection is an open circular ring centered at z_0 . Figure illustrates such a ring centered at the origin. The set defined by (2) is called an open circular annulus. By allowing $\rho_1 = 0$, we obtain a deleted neighborhood of z_0 .

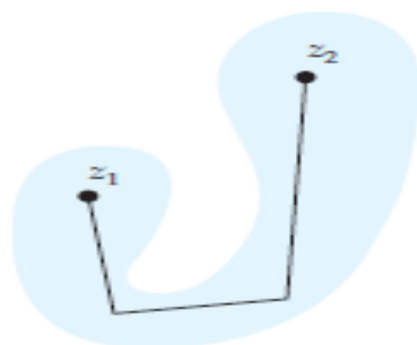


$1 < |z| < 2$; interior of circular ring

Domain

If any pair of points z_1 and z_2 in a set S can be connected by a polygonal line that consists of a finite number of line segments joined end to end that lies entirely in the set, then the set S is said to be connected.

An open connected set is called a domain.



Connected set

Regions A region is a set of points in the complex plane with all, some, or none of its boundary points. Since an open set does not contain any boundary points, it is automatically a region. A region that contains all its boundary points is said to be **closed**. The disk defined by $|z - z_0| \leq \rho$ is an example of a closed region and is referred to as a **closed disk**. A neighborhood of a point z_0 defined by $|z - z_0| < \rho$ is an open set or an open region and is said to be an **open disk**. If the center z_0 is deleted from either a closed disk

Bounded Sets Finally, we say that a set S in the complex plane is **bounded** if there exists a real number $R > 0$ such that $|z| < R$ every z in S . That is, S is bounded if it can be completely enclosed within *some* neighborhood of the origin. In Figure 1.1, the set S shown in color is bounded because it is contained entirely within the dashed circular neighborhood of the origin. A set is **unbounded** if it is not bounded. For example, the set in Figure 1.1(d) is bounded, whereas the sets in Figures 1.1(a), 1.1(b), and 1.1(c) are unbounded.

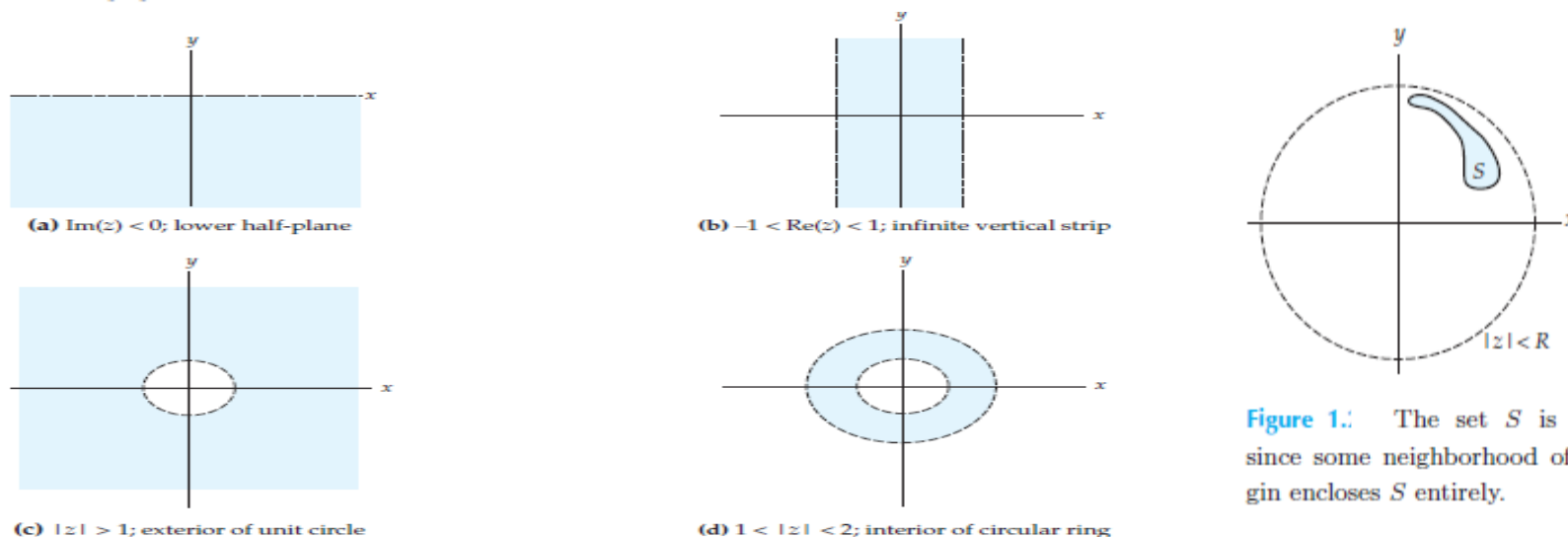
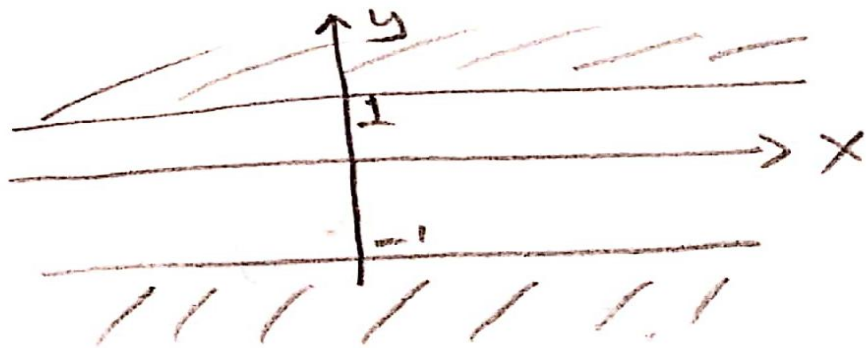


Figure 1.1 The set S is bounded since some neighborhood of the origin encloses S entirely.

S: $\{ | \operatorname{Im} z | > 1 \}$ is it connected?

$$\operatorname{Im} z < -1 \quad \operatorname{Im} z > 1$$

$$\operatorname{Im} z = y \quad y > 1 \quad y < -1$$



it is not
connected

whether $|z-4| \geq |z|$ shows a domain!

$$|z-4| \geq |z|$$

$$|x+iy-4| \geq |x+iy|$$

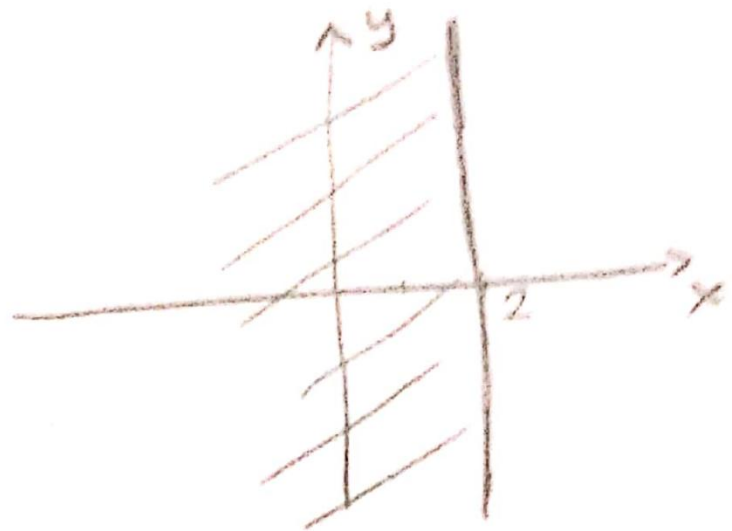
$$\sqrt{(x-4)^2 + y^2} \geq \sqrt{x^2 + y^2}$$

$$(x-4)^2 + y^2 \geq x^2 + y^2$$

$$\cancel{x^2} - 8x + 16 + \cancel{y^2} \geq \cancel{x^2} + \cancel{y^2}$$

$$16 \geq 8x$$

$$2 \geq x$$



It is not open
(It is connected)

it is not domain

additional information on definitions

By a **point set** in the complex plane we mean any sort of collection of finitely many or infinitely many points. Examples are the solutions of a quadratic equation, the points of a line, the points in the interior of a circle as well as the sets discussed just before.

A set S is called **open** if every point of S has a neighborhood consisting entirely of points that belong to S . For example, the points in the interior of a circle or a square form an open set, and so do the points of the right half-plane $\operatorname{Re} z = x > 0$.

A set S is called **connected** if any two of its points can be joined by a chain of finitely many straight-line segments all of whose points belong to S . An open and connected set is called a **domain**. Thus an open disk and an open annulus are domains. An open square with a diagonal removed is not a domain since this set is not connected. (Why)

The **complement** of a set S in the complex plane is the set of all points of the complex plane that *do not* belong to S . A set S is called **closed** if its complement is open. For example, the points on and inside the unit circle form a closed set ("closed unit disk") since its complement $|z| > 1$ is open.

A **boundary point** of a set S is a point every neighborhood of which contains both points that belong to S and points that do not belong to S . For example, the boundary points of an annulus are the points on the two bounding circles. Clearly, if a set S is open, then no boundary point belongs to S ; if S is closed, then every boundary point belongs to S . The set of all boundary points of a set S is called the **boundary** of S .

A **region** is a set consisting of a domain plus, perhaps, some or all of its boundary points. **WARNING** "Domain" is the *modern* term for an open connected set. Nevertheless, some authors still call a domain a "region" and others make no distinction between the two terms.

For example, in a course in calculus you dealt with *limits at infinity*, where the behavior of functions was examined as x either increased or decreased without bound. Since we have exactly two directions on a number line, it is convenient to represent the notions of “increasing without bound” and “decreasing without bound” symbolically by $x \rightarrow +\infty$ and $x \rightarrow -\infty$, respectively. It turns out that we can get along just fine without the designation $\pm\infty$ by dealing with an “ideal point” called the **point at infinity**, which is denoted simply by ∞ . To do this we identify any real number a with a point (x_0, y_0) on a unit circle $x^2 + y^2 = 1$ by drawing a straight line from the point $(a, 0)$ on the x -axis or horizontal number line to the point $(0, 1)$ on the circle. The point (x_0, y_0) is the point of intersection of the line with the circle.

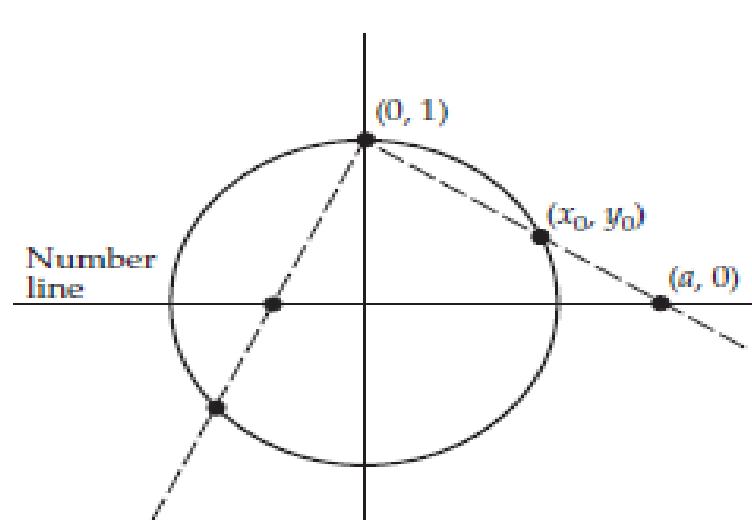
The only point on the circle that does not correspond to a real number a is $(0, 1)$. To complete the correspondence with *all* the points on the circle, we identify $(0, 1)$ with ∞ . The set consisting of the real numbers $\mathbf{R}$ adjoined with ∞ is called the **extended real-number system**.

Recall that since $\mathbf{C}$ is not ordered, the notions of z either “increasing” or “decreasing” have no meaning. However, we know that by increasing the modulus $|z|$ of a complex number z , the number moves farther from the origin. If we allow $|z|$ to become unbounded, say, along the real and imaginary axes, we do not have to distinguish between “directions” on these axes by notations such as $z \rightarrow +\infty$, $z \rightarrow -\infty$, $z \rightarrow +i\infty$, and $z \rightarrow -i\infty$. In complex analysis, only the notion of ∞ is used because we can extend the complex number system $\mathbf{C}$ in a manner analogous to that just described for the real number system $\mathbf{R}$. This time, however, we associate a complex number with a point on a unit sphere called the **Riemann sphere**. By drawing a line from the number $z = a + ib$,

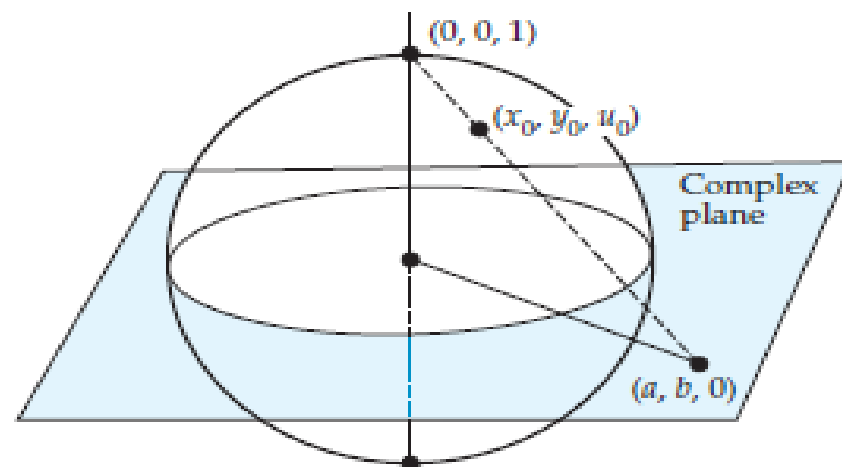
written as $(a, b, 0)$, in the complex plane to the north pole $(0, 0, 1)$ of the sphere $x^2 + y^2 + u^2 = 1$, we determine a unique point (x_0, y_0, u_0) on a unit sphere. a complex number with a very large modulus is far from the origin $(0, 0, 0)$ and, correspondingly, the point (x_0, y_0, u_0) is close to $(0, 0, 1)$. In this manner each complex number is identified with a single point on the sphere.

Because the point $(0, 0, 1)$ corresponds to no number z in the plane, we correspond it with ∞ . Of course, the system consisting of $\mathbb{C}$ adjoined with the “ideal point” ∞ is called the **extended complex-number system**.

This way of corresponding or mapping the complex numbers onto a sphere—north pole $(0, 0, 1)$ excluded—is called a **stereographic projection**.



(a) Unit circle



(b) Unit sphere

Solve the quadratic equation $z^2 + (1 - i)z - 3i = 0$.

Solution With $a = 1$, $b = 1 - i$, and $c = -3i$ we have

$$z = \frac{-(1 - i) + [(1 - i)^2 - 4(-3i)]^{1/2}}{2} = \frac{1}{2} \left[-1 + i + (10i)^{1/2} \right]. \quad ($$

To compute $(10i)^{1/2}$ we write $10i = \sqrt{10} e^{i\pi/2}$, $r = \sqrt{10}$, $\theta = \pi/2$, and $n = 2$, $k = 0$, $k = 1$. The two square roots of $10i$ are:

$$w_0 = \sqrt{10} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = \sqrt{10} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right) = \sqrt{5} + \sqrt{5}i$$

and

$$w_1 = \sqrt{10} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) = \sqrt{10} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right) = -\sqrt{5} - \sqrt{5}i.$$

Therefore,

$$z_1 = \frac{1}{2} \left[-1 + i + \left(\sqrt{5} + \sqrt{5}i \right) \right] \quad \text{and} \quad z_2 = \frac{1}{2} \left[-1 + i + \left(-\sqrt{5} - \sqrt{5}i \right) \right].$$

These solutions of the original equation, written in the form $z = a + ib$, are

$$z_1 = \frac{1}{2} \left(\sqrt{5} - 1 \right) + \frac{1}{2} \left(\sqrt{5} + 1 \right)i \quad \text{and} \quad z_2 = -\frac{1}{2} \left(\sqrt{5} + 1 \right) - \frac{1}{2} \left(\sqrt{5} - 1 \right)i.$$

$$e^z = e^{x+iy} = e^x(\cos y + i \sin y).$$

$$e^{z_1}e^{z_2} = e^{z_1+z_2}.$$

$$z = re^{i\theta}.$$

This convenient form is called the **exponential form** of a complex number z . For example, $i = e^{\pi i/2}$ and $1 + i = \sqrt{2}e^{\pi i/4}$. Also, the formula for the n n th roots of a complex number,

$$z^{1/n} = \sqrt[n]{r}e^{i(\theta+2k\pi)/n}, k = 0, 1, 2, \dots, n-1.$$